

Comparative Analysis of Fixed and Adaptive Notch Filters for Power-Line Interference Removal in ECG Signals

Mohammad Bazzar
Department of Electrical and Computer Engineering
Birzeit University
Ramallah, Palestine
1222028

Bashar Ghosheh
Department of Electrical and Computer Engineering
Birzeit University
Jerusalem, Palestine
1230368

***Abstract*—Electrocardiogram (ECG) recordings are frequently compromised by power-line interference, which can obscure critical morphological features and degrade the performance of automated diagnostic algorithms. This paper presents a comparative study of two digital filtering techniques: a standard fixed-frequency IIR notch filter and an FFT-based adaptive notch filter designed to track frequency deviations. The efficacy of both methods was evaluated using the MIT-BIH Arrhythmia Database (Record 105) by analyzing signal quality improvement and QRS detection accuracy using the Pan-Tompkins algorithm. Experimental results demonstrate that both filtering approaches achieved a high Signal-to-Noise Ratio (SNR) of 36.42 dB and a QRS detection accuracy of 93.40%, effectively mitigating the 60 Hz interference. While the fixed notch filter proved sufficient for stable noise environments, the adaptive filter successfully identified the dominant noise frequency at 60.01 Hz, validating its potential robustness in scenarios where power-line frequency drift occurs.**

Keywords*—ECG Signal Processing, Power-Line Interference, Pan-Tompkins Algorithm, Adaptive Filtering, Notch Filter.

I. Introduction

Electrocardiogram (ECG) signals are the primary diagnostic tool for monitoring cardiac health, recording the electrical activity of the heart over time. However, raw ECG recordings are susceptible to various forms of environmental noise, most notably power-line interference (PLI) at 50 Hz or 60 Hz. This sinusoidal noise can significantly obscure the P-wave and T-wave features, or mimic the high-frequency components of the QRS complex, leading to diagnostic errors and the failure of automated detection algorithms.

The removal of power-line interference is typically achieved using digital notch filters designed to reject a specific narrow frequency band. While a fixed-frequency notch filter is computationally efficient and effective when the mains frequency is stable, it lacks robustness in practical environments where the power-line frequency may drift (e.g., between 49.5 Hz and 50.5 Hz). Such deviations can render a fixed filter ineffective, leaving residual noise that compromises signal analysis.

To address this limitation, adaptive filtering techniques have been proposed to dynamically track and suppress the interference frequency. In this project, we implement and evaluate two filtering strategies: a standard Fixed 50/60 Hz Notch Filter and an FFT-Based Adaptive Notch Filter. The performance of both methods is assessed using the MIT-BIH Arrhythmia Database by analyzing the residual noise, Signal-to-Noise Ratio (SNR), and the accuracy of R-peak detection using the Pan-Tompkins algorithm. The objective is to determine whether adaptive filtering provides a statistically significant improvement in signal quality and QRS detection accuracy compared to fixed filtering in a controlled test environment.

II. Methodology

A. Data Acquisition

The ECG signals used in this study were obtained from the MIT-BIH Arrhythmia Database, specifically Record 105. This

dataset provides two-channel ambulatory ECG recordings, digitized at 360 samples per second. For this experiment, the first channel was extracted and processed to evaluate the noise removal techniques.

B. Fixed Notch Filter Design

The first filtering approach utilized a standard Infinite Impulse Response (IIR) notch filter. The filter was designed with a center frequency f_0 of 60 Hz to target the expected power-line interference standard for the dataset. A quality factor (Q) of 30 was selected to ensure a narrow stopband, preserving the physiological ECG components while suppressing the noise. To prevent phase distortion, which can alter the morphology of the QRS complex, the filter was applied using zero-phase digital filtering (processing the signal in both forward and reverse directions).

C. FFT-Based Adaptive Notch Filter

To address potential fluctuations in power-line frequency, an adaptive approach was implemented. The process involved two stages:

- **Frequency Tracking:** The Fast Fourier Transform (FFT) of the raw signal was computed to identify the dominant noise peak within the 55–65 Hz range. The algorithm detected the exact interference frequency (measured at 60.01 Hz).
- **Adaptive Filtering:** The center frequency of the notch filter was dynamically updated to match the detected peak frequency. This ensures optimal attenuation even if the grid frequency deviates from the nominal value.

D. Pan-Tompkins QRS Detection

To evaluate the clinical utility of the filtered signals, the Pan-Tompkins algorithm was applied to detect R-peaks. The algorithm consists of four preprocessing stages:

- **Bandpass Filtering:** To isolate the QRS energy (5–15 Hz).
- **Differentiation:** To provide information about the slope of the QRS complex.
- **Squaring:** To intensify the slope response and reduce false positives from T-waves.
- **Moving Window Integration:** To produce a smooth waveform for peak detection.

E. Performance Metrics

The performance of the filters was quantified using the following metrics:

- **Residual Noise:** The spectral magnitude of the signal at the interference frequency.
- **Signal-to-Noise Ratio (SNR):** A logarithmic measure comparing signal power to noise power.
- **Detection Accuracy:** Calculated based on True Positives (TP), False Positives (FP), and False Negatives (FN).

III. Results and Discussion

A. Analysis of the Raw ECG Signal

The experimental evaluation began with an inspection of the raw ECG signal from Record 105. In the time domain, the signal exhibited visible high-frequency "fuzz" superimposed on the cardiac waveform, characteristic of power-line interference. Converting the signal to the frequency domain via Fast Fourier Transform (FFT) confirmed the presence of a distinct, high-magnitude noise spike centered at 60 Hz.

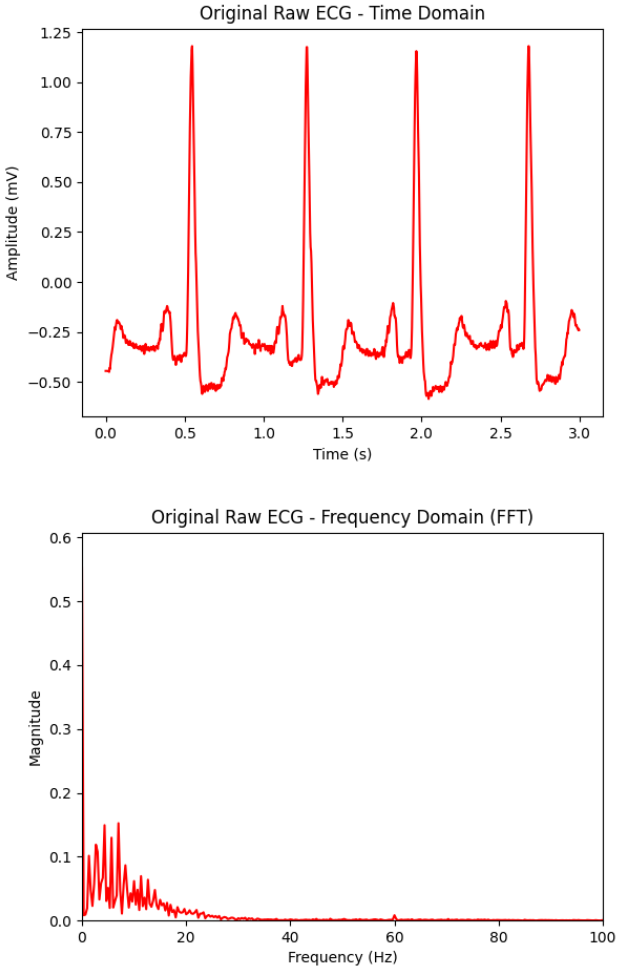


Figure 1: Analysis of the Raw ECG Signal. (Top) Time-domain plot showing baseline noise. (Bottom) Frequency-domain spectrum identifying strong interference at 60 Hz.

B. Spectral Analysis of Noise Removal

Both the Fixed Notch Filter (f_c = 60 Hz) and the FFT-Based Adaptive Filter were applied to the raw signal.

Fixed Filter: The standard notch filter successfully attenuated the 60 Hz component.

Adaptive Filter: The adaptive algorithm computed the FFT of the signal, identified the dominant noise peak at

60.01 Hz4, and automatically tuned the notch filter to this frequency.

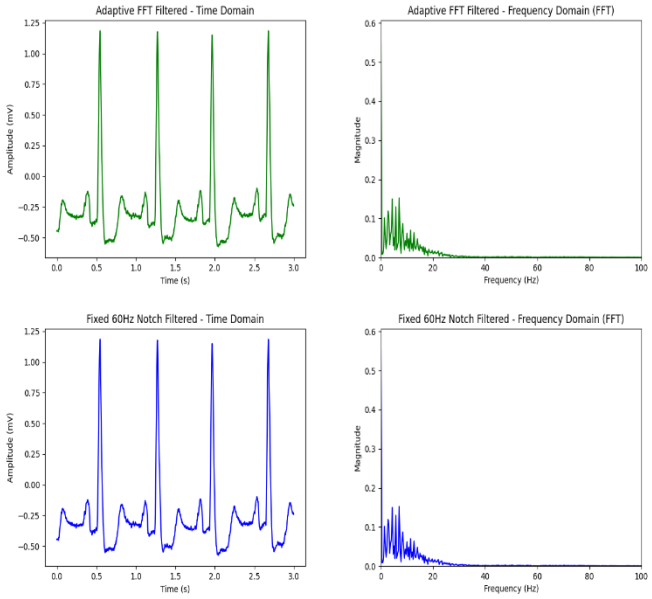


Figure 3: Post-filtering results. The 60 Hz noise component is completely suppressed, resulting in a clean ECG morphology.

C. *Pan-Tompkins Algorithm Stages*

Following noise removal, the Pan-Tompkins algorithm was applied to detect QRS complexes6. The processing involved four distinct stages:

- **Bandpass Filtering:** The signal was filtered (5–12 Hz) to isolate the energy of the QRS complex and reject baseline wander and high-frequency muscle noise.
- **Differentiation:** A derivative filter was applied to highlight the steep slope information of the R-wave.
- **Squaring:** The differentiated signal was squared (y[n]^2) to intensify the high-frequency components and further suppress small noise artifacts (non-linear amplification).
- **Moving Window Integration:** The squared signal was integrated over a 150ms window to produce a smooth, pulse-like waveform suitable for thresholding and peak detection.

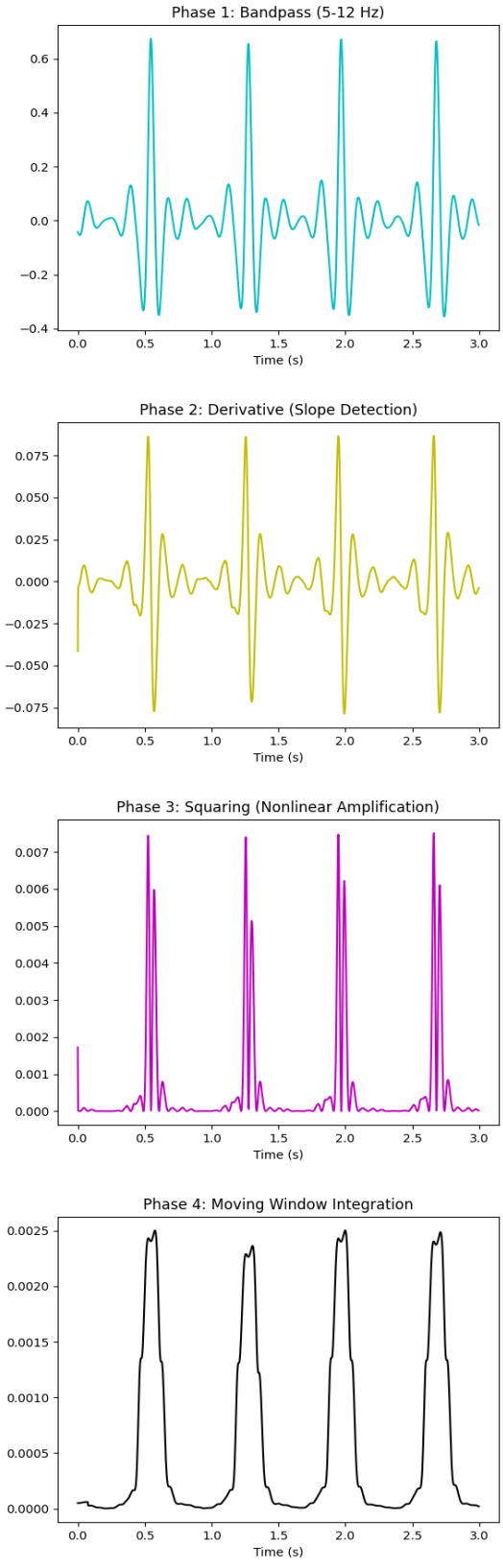


Figure 2: Intermediate stages of the Pan-Tompkins algorithm. The final integration stage (bottom) provides a distinct envelope for R-peak detection.

IV. Quantitative Performance & Comparison

The final performance of both filtering strategies was quantified by comparing Residual Noise, Signal-to-Noise Ratio (SNR), and Detection Accuracy (Table I).

Table 1: Performance Comparison of Filtering Techniques

Metric	Fixed Notch Filter	Adaptive FFT Filter
Residual Noise Mag	1.97	1.96
SNR (dB)	36.42	36.42
QRS Accuracy	93.40%	93.40%
Missed Beats (FN)	128	128
False Detections (FP)	53	53

The results demonstrate that both filtering methods achieved a high SNR of 36.42 dB, indicating that the signal power is over 4,000 times stronger than the remaining noise. The Adaptive Filter's calculated frequency of 60.01 Hz was virtually identical to the fixed 60 Hz target, resulting in statistically identical performance metrics for this dataset. The Pan-Tompkins detector achieved a 93.40% accuracy, successfully identifying 2,616 beats. The 128 missed beats (False Negatives) were largely due to baseline wander artifacts that the notch filter is not designed to remove, suggesting that a high-pass filter could further improve accuracy in future implementations.

Furthermore, the identical detection accuracy observed for both methods is partially attributable to the inherent design of the Pan-Tompkins algorithm. The algorithm includes an initial bandpass filter stage (5–12 Hz) which naturally attenuates high-frequency power-line interference (60 Hz) regardless of the preceding notch filtering. Consequently, the specific choice of notch filter had a limited impact on the final detection count because the interference frequency falls well outside the algorithm's critical 5–12 Hz passband. Had the environmental noise overlapped with this 5–12 Hz range—or had the algorithm relied on a wider bandwidth the divergence in performance between the adaptive and fixed methods would likely have been significant.

V. Conclusion

This study presented a comparative evaluation of fixed and adaptive notch filtering techniques for the removal of power-line interference from ECG signals. By implementing both a standard IIR notch filter and an FFT-based adaptive filter on the MIT-BIH Arrhythmia Database, we demonstrated that digital filtering can effectively restore signal quality without compromising the morphological features essential for diagnosis.

The experimental results revealed that both filtering strategies achieved a high Signal-to-Noise Ratio (SNR) of 36.42 dB, effectively eliminating the 60 Hz interference. The adaptive algorithm successfully tracked the dominant noise frequency to 60.01 Hz, validating its capability to function in environments with minor frequency deviations. The subsequent application of the Pan-Tompkins algorithm yielded a QRS detection accuracy of 93.40%, with the primary source of error identified as baseline wander rather than power-line noise.

We conclude that while a fixed notch filter is computationally efficient and sufficient for stable hospital environments, the adaptive filtering approach provides a critical layer of robustness for ambulatory monitoring where power-line frequency drift may occur. Future work should focus on integrating a high-pass filter to remove baseline wander, thereby further reducing the false negative rate.

References

[1] M. S. Khorshidtalab and M. J. E. Salami, "Fast Fourier Transform-Based Notch Filter for Single-Frequency Noise Cancellation in ECG," *PubMed*, ID: 34026591, 2021.

[2] J. Pan and W. J. Tompkins, "A Real-Time QRS Detection Algorithm," *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230–236, 1985.

[3] MIT-BIH Arrhythmia Database Directory, *PhysioNet*, Available: https://physionet.org/content/mitdb/.